Contact

kzm6@cornell.edu

www.linkedin.com/in/khojasteh-mirza (LinkedIn)

Top Skills

Analytical Skills

Project Management

Computer Vision

Languages

Hindi (Native or Bilingual)

German (Elementary)

English (Full Professional)

Certifications

Business English for Cross-cultural Communication

Python for Data Science

Certificate of Appreciation (Top 3) - IWEM 2020 - Mathematical Modeling and Prediction

Speak English Professionally: In Person, Online & On the Phone

Introduction to Self-Driving Cars - University of Toronto

Publications

Reviewing the Scope of Triboelectric Nanogenerators as a feasible solution and its applications

Linear Quadratic Gaussian Regulator for the Nonlinear Observer-Based Control of a Dynamic Base Inverted Pendulum

A Study on Autonomous Mechanisms for Swapping of Batteries on Unmanned Aerial Vehicles

Design of Mobile Docking Mechanism for Unmanned Aerial Vehicles capable of Vertical Take-off and Landing

Khojasteh Mirza

Data Science | Associate Research Scientist - Mount Sinai Neuroscience | Machine Learning

New York, New York, United States

Summary

I am a Data Scientist building biologically inspired artificial neural networks. Currently, I am working as an Associate Researcher in the rapidly evolving field of Machine Learning and Computational Neuroscience at Mount Sinai in NYC. I am passionate about applying AI and data science to solve real-world problems, especially in the healthcare domain. I also have a Master's degree in Electrical and Computer Engineering from Cornell University.

Previously, I worked as a Graduate Research Assistant at the Lab for Intelligent Systems and Control, where I developed AI algorithms for Ferrari's driver-assist systems, leveraging sensor fusion and data science. I also collaborated with Tata Consultancy Services to develop a system capable of detecting Parkinson's Disease before symptoms appear, using data analysis and machine learning.

Before that, I worked as a Research Analyst at AIRMO, where I built Vision Transformer modules for automated methane detection on satellites. I conducted in-depth data analysis using NASA Emit and Sentinel-5P data to identify the worst contributors to methane emissions, providing valuable insights for emission reduction efforts.

I hold a bachelor's degree in Mechatronics from NMIMS Mumbai, India. After graduation, I worked as a Researcher at IIT-Bombay's Aerospace Department, where I contributed to enhancing surveillance systems by optimizing battery-swapping processes. This innovation extended UAV flight times from 60 minutes to 30 days, and I led a cross-functional team to implement these improvements.

I have multiple publications, certifications, and skills in Robotics, Data Science, and AI. I am proficient in Pytorch, Python, SQL, and Tableau.

A Review of Optimization of Morphing Surfaces for Increased Aerodynamic Performance

I am a curious, analytical, and creative thinker who thrives on challenging and impactful projects. My goal is to use AI and data science to develop innovative solutions that improve lives and benefit the environment.

Experience

The Friedman Brain Institute
Associate Machine Learning Researcher
June 2024 - Present (2 years 2 months)
New York City Metropolitan Area

• Developing artificial neural networks that are inspired by biological neural networks.
• Building Embodied Transfer Learning - (RL + Transfer Learn + Control Systems) which allows networks to learn faster (15-20%), and learn more (10%). A GRU learning a specific task can inform and improve performance on diverse tasks.
• Also building deep neural networks for mimicking limb movements (proprioception).

Cornell University
Graduate Research Assistant
September 2023 - January 2024 (5 months)
Ithaca, New York, United States

LISC - Lab for Intelligent Systems and Control

• Worked on a project for Ferrari S.p.A to create AI algorithms for Advanced Driver Assisted Systems (ADAS) using data science and sensor fusion.
• Using MATLAB to create and simulate driving scenarios, and gathered simulation data through cameras and radars.
• Leveraged Pandas and NumPy for data manipulation and sensor fusion for monitoring opposing lane traffic to predict valid overtaking opportunities using Frenet Frames.

AIRMO
Data Scientist
June 2023 - August 2023 (3 months)
Berlin, Germany

• Built a vision transformer module to detect methane plumes from Sentinel data

• Conducted in-depth data analysis using NASA Emit and Sentinel-5P data to identify methane emissions sources.
• Discovered agriculture accounts for over 75% of methane emissions, emphasizing the need for targeted mitigation strategies.
• Utilized machine learning to reduce uncertainty in methane emissions detection by accounting for cloud cover and aerosol effects.

Indian Institute of Technology, Bombay
2 years 1 month

Researcher
October 2022 - July 2023 (10 months)
Mumbai, Maharashtra, India

• Enhanced surveillance systems, extending UAV flight duration from 60 minutes to 30 days.
• Created a mid-air battery swapping process, improving flight time and drone productivity.
• Led a cross-functional team to ensure seamless integration of robotics systems.

Research Intern
July 2021 - September 2022 (1 year 3 months)
Mumbai, Maharashtra, India

Ozibook
Data Science Intern
May 2023 - June 2023 (2 months)
Bengaluru, Karnataka, India

• Collected and organized data from LinkedIn using Python, creating a list of 1500 high-engagement users for targeted marketing.
• Conducted A/B tests to optimize content types and enhance user engagement.
• Utilized Tableau to visualize engagement trends and demographics, facilitating data-driven decisions for sales and marketing teams.

Baisco
Project Intern
August 2022 - October 2022 (3 months)
Bengaluru, Karnataka, India

• Conducted research to enhance control system stability by 15% through dihedral angles on hexacopter chassis.

• Implemented manufacturing tweaks for 3D printing, reducing overhangs and saving time.

Technotix MPSTME

2 years 1 month

Head - Design and Simulation

September 2020 - September 2021 (1 year 1 month)

Mumbai, Maharashtra, India

• Led a team of 10 in designing 2 autonomous robots for ABU Robocon competition, achieving 100/100 in the Design Stage.

• Secured an impressive All India Rank 11 in the competition's finals.

• Developed skills in Solidworks, project management, and team leadership.

Subhead - Design and Simulation

September 2019 - September 2020 (1 year 1 month)

Mumbai, Maharashtra, India

• Designed autonomous robots in Solidworks and simulated them in MATLAB using Simscape Multibody.

• Achieved All India rank 18 in a national competition.

• One of only two teams that year to have a running simulation.

Pawan Hans

Research Intern

May 2021 - June 2021 (2 months)

Mumbai, Maharashtra, India

• Developed optimal control system simulation for the Airbus AS-365N and N3 variants, focusing on yawing motion and torque of main rotors.

• Formulated Distributed Full Authority Digital Engine Control architecture to enhance engine performance and efficiency.

• Tested and maintained various communication systems and electronic components for Pawan Hans in Mumbai, India.

Decibels Lab

Internship

November 2020 - December 2020 (2 months)

Greater Bengaluru Area

• Conducted mathematical modeling of vehicle resistive forces on a Formula 1 vehicle in MATLAB.

• Developed a vehicle dynamics simulator with a Numerical G-G Diagram for parameter exploration.

• Collaborated remotely with Decibels Lab in the Greater Bengaluru Area to optimize chassis conditions.

SVKM's NMIMS Mukesh Patel School of Technology Management & Engineering
Undergraduate Researcher
May 2020 - June 2020 (2 months)
Mumbai, Maharashtra, India

• Developed a fuzzy logic controller for MPPT to optimize energy for the load grid.
• Designed a 3-phase inverter circuit for converting DC to 3-phase AC efficiently.
• Repurposed the system for batteries, maximizing power output for charging.

Education

Cornell University
Master of Engineering - MEng, Computer Engineering · (July 2023 - May 2024)

Cornell Tech
Master of Engineering - MEng, Computer Engineering · (July 2023 - May 2024)

SVKM's Narsee Monjee Institute of Management Studies (NMIMS)
Bachelor of Technology - B.Tech, Honors: Automation in Automobile and Manufacturing Processes , Mechatronics, Robotics, and Automation Engineering · (2018 - 2022)